Comp maths Sheet 1b

(3) a) $a_n = 2^{-n}$

$$\left|\frac{a_{n+1}}{a_n}\right| = \frac{2^{-(n+1)}}{2^{-n}} = \frac{1}{2}$$

$\therefore$ order of convergence $= 1$.

b) $a_n = (n+1)^{-1}$

$$\left|\frac{a_{n+1}}{a_n}\right| = \frac{n+1}{n+2} \longrightarrow 1$$

$\therefore$ order of convergence $= 1$

c) $a_n = 2^{-2^n}$

$$a_{n+1} = 2^{-2^{n+1}} = 2^{(-2^n)\times 2} = \left(2^{-2^n}\right)^2 = (a_n)^2$$

$\therefore$ order of convergance $= 2$.

(5) a) $\dfrac{e_{n+1}}{(e_n)^q} \approx M \approx \dfrac{e_{n+2}}{(e_{n+1})^q}$

$$\left(\frac{e_{n+1}}{e_n}\right)^q \approx \frac{e_{n+2}}{e_{n+1}}$$

$$q \approx \frac{\ln(e_{n+2}) - \ln(e_{n+1})}{\ln(e_{n+1}) - \ln(e_n)}$$

$$q = \frac{\ln(e_{n+1}) - \ln M}{\ln(e_n)} = \frac{\ln(e_{n+2}) - \ln M}{\ln(e_{n+1})}$$

$$\Rightarrow M = \exp\left(\frac{\ln(e_{n+1})^2 - \ln(e_{n+2})\ln(e_n)}{\ln(e_{n+1}) - \ln(e_n)}\right)$$